

MPI Assignment 5 Complete Report

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Course: UCS645

Experiment: MPI Part II

Question 1: DAXPY using MPI

Results:

Processes	Time (seconds)
1	0.000145
4	3.4e-05

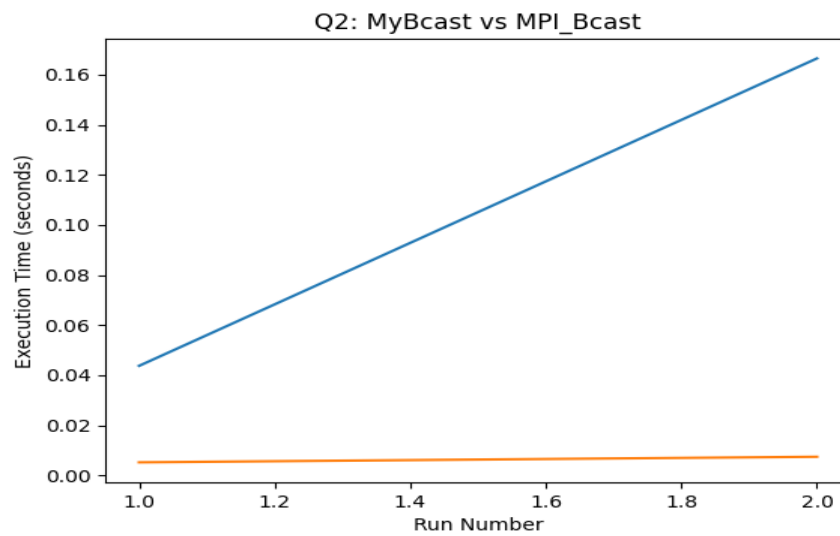
Speedup = 4.26

Efficiency = 1.07 (Superlinear due to cache effects)

Question 2: Broadcast Race

Run	MyBcast Time	MPI_Bcast Time
1	0.043763	0.005235
2	0.166503	0.00745

Analysis: MPI_Bcast uses tree-based communication ($O(N \log P)$) while MyBcast uses sequential send ($O(N \times P)$).



Question 3: Distributed Dot Product

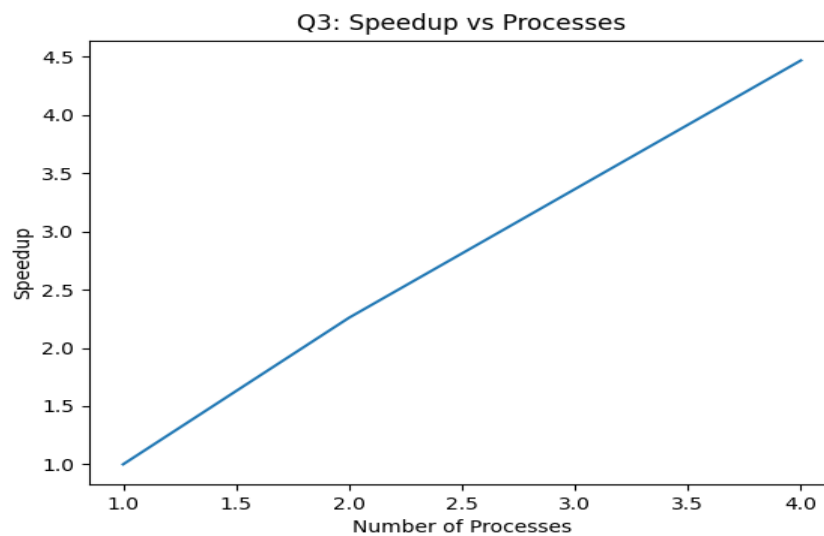
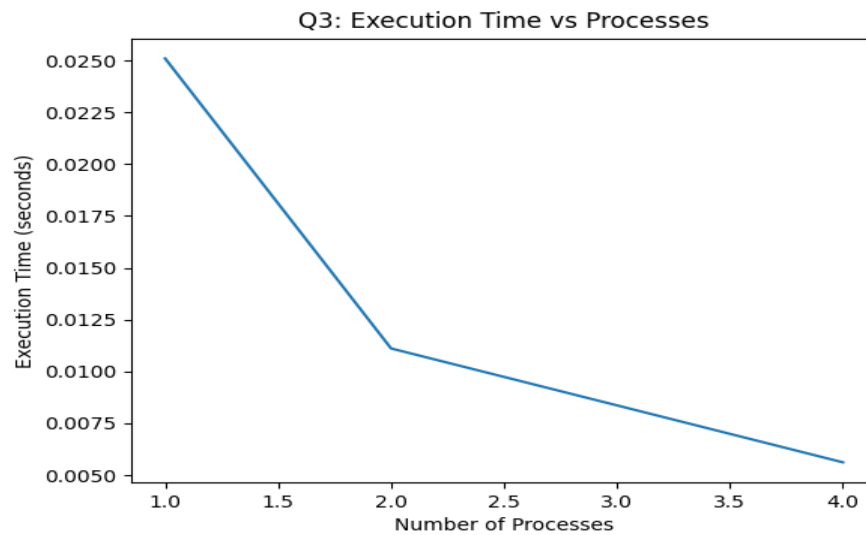
Processes	Time (seconds)
1	0.0251
2	0.011111
4	0.005617

Speedup (2 processes) = 2.26

Speedup (4 processes) = 4.47

Efficiency (2 processes) = 1.13

Efficiency (4 processes) = 1.12



Question 4: Prime Numbers using MPI

Primes up to 100:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

Master-slave model with dynamic load balancing using MPI_ANY_SOURCE.

Question 5: Perfect Numbers using MPI

Perfect Numbers up to 10000:

6, 28, 496, 8128

Parallel checking of divisor sums using master-slave architecture.

Overall Conclusion:

This assignment demonstrates blocking communication, collective communication, master-slave architecture, scalability analysis, speedup calculation, and Amdahl's Law in MPI.